



I. INTRODUCTION

The focus of this project is to develop an application that uses computer vision to serve as an automated personal trainer. The application allows users to perform exercises using their smartphone camera to detect body movements, automatically count repetitions, and receive real-time feedback on their posture and form. The system will utilize on-device pose estimation to map body landmarks, ensuring effective home workouts without the need for continuous internet connectivity or specialized gym equipment.

The initial version will provide core features, including a real-time camera interface for pose detection, automatic repetition counting for specific exercises (e.g., Squats, Jumping Jacks), instant visual alerts for form correction, and a local activity log to track daily progress.

The application will generate outputs, including verified workout sessions, repetition accuracy statistics, and a visual activity consistency grid. It does not account for extremely low-light environments, scenarios where the user's body is fully occluded from the camera's view, or complex compound exercises that fall outside the pre-programmed library.

II. FUNCTIONAL SPECIFICATION

2.1 Mock-Up

The system interface will feature user-friendly and accessible designs to facilitate seamless interaction for all stakeholders. Sample mock-ups to be developed and included in this document are as follows:

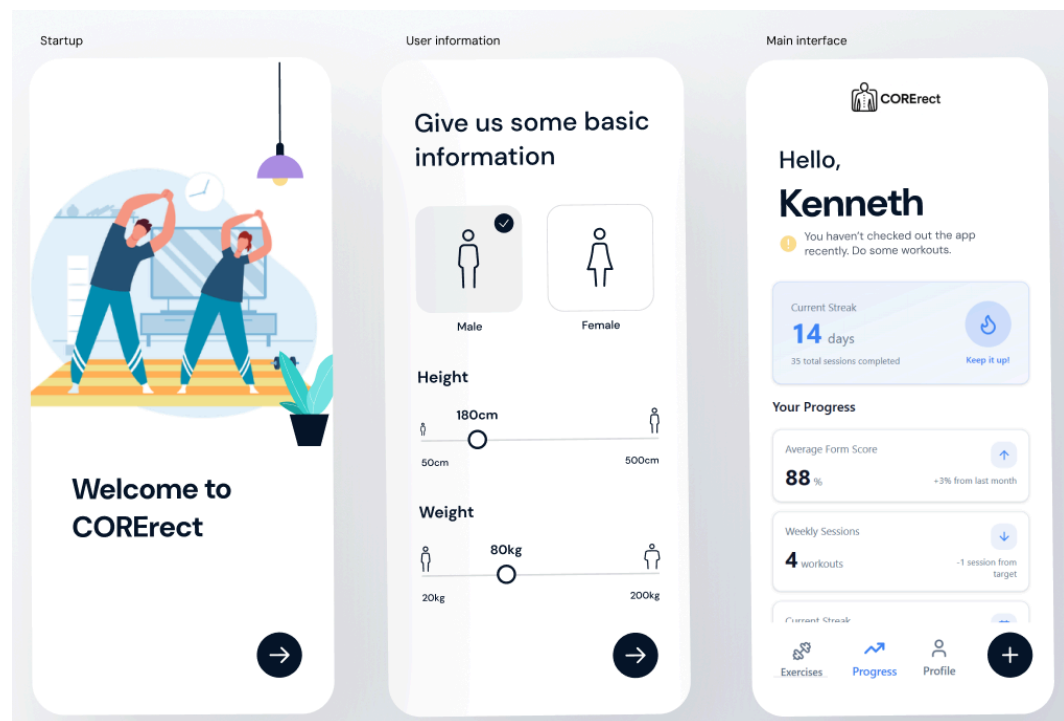


Figure 1: Main Interface of CORErect

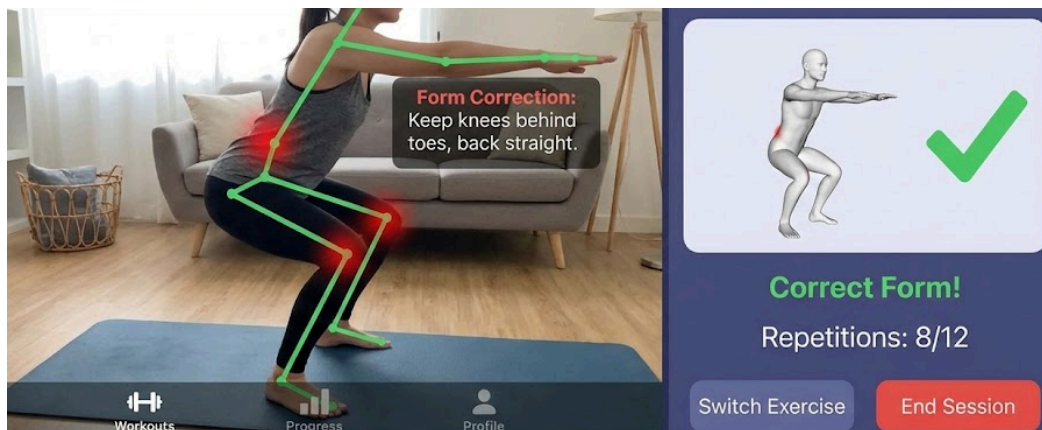


Figure 2: CORErect Live Exercise Session

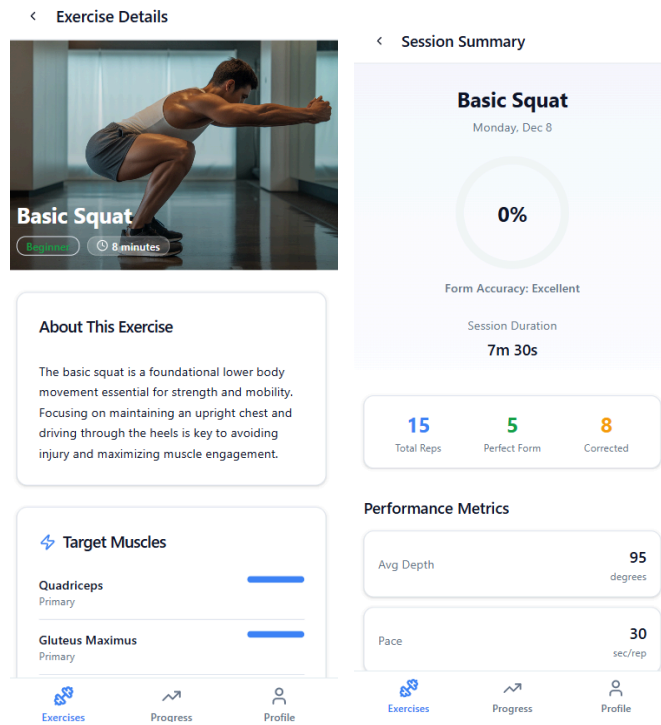


Figure 3: Exercise Details and Summary

2.2 Functional Requirements

- a. **Real-Time Pose Detection:** The system uses the device's camera to identify and track the user's body landmarks in real-time. It draws a digital skeleton overlay on the screen to confirm detection.
- b. **Automated Repetition Counting:** The system algorithmically determines when a repetition is complete based on body angles.
 - Squat: The counter increments only when the user's hip landmark descends below the knee landmark and returns to a standing position.
- c. **Form Correction Feedback:** The system shall provide immediate feedback if specific geometric constraints are violated.
 - Condition: If the user's back angle deviates more than 15 degrees from vertical during a squat.
 - Action: Display text "Keep Back Straight" and provide an audio cue.
- d. **Activity Consistency Grid:** The system shall display a calendar-style grid where days with completed workouts are colored to encourage daily streaks.
- e. **Workout History Log:** The system shall store details of every session (Date, Exercise Type, Total Reps) in the local database for user review.